is the commutator of the matrices

$$A = \begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

while the matrix

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$$

is the commutator of the matrices

$$A = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This completes the proof. □

Therefore,

$$G/[G,G] \cong \mathbb{F}_q^\times \quad \text{via } g \mapsto \det(g).$$

The 1-dimensional representations of G thus have the form

$$\rho(g) = \xi\big(\det(g)\big),$$

where ξ is a homomorphism

$$\xi : \mathbb{F}_q^\times \to \mathbb{C}^\times;$$

so there are $q-1$ such representations, denoted $\mathbb{C}_\xi$.

5.25.3. Principal series representations. Let

$$B \subset G, \quad B = \{\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}\}$$

(the set of upper triangular matrices); then

$$|B| = (q-1)^2 q,$$

$$[B,B] = U = \{\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}\},$$

and

$$B/[B,B] \cong \mathbb{F}_q^\times \times \mathbb{F}_q^\times$$

(the isomorphism maps an element of B to its two diagonal entries). Let

$$\lambda : B \to \mathbb{C}^\times$$